

Probabilistic hyperbolic embedding of networks

Combining network geometry with Bayesian inference

Ariane Lizotte

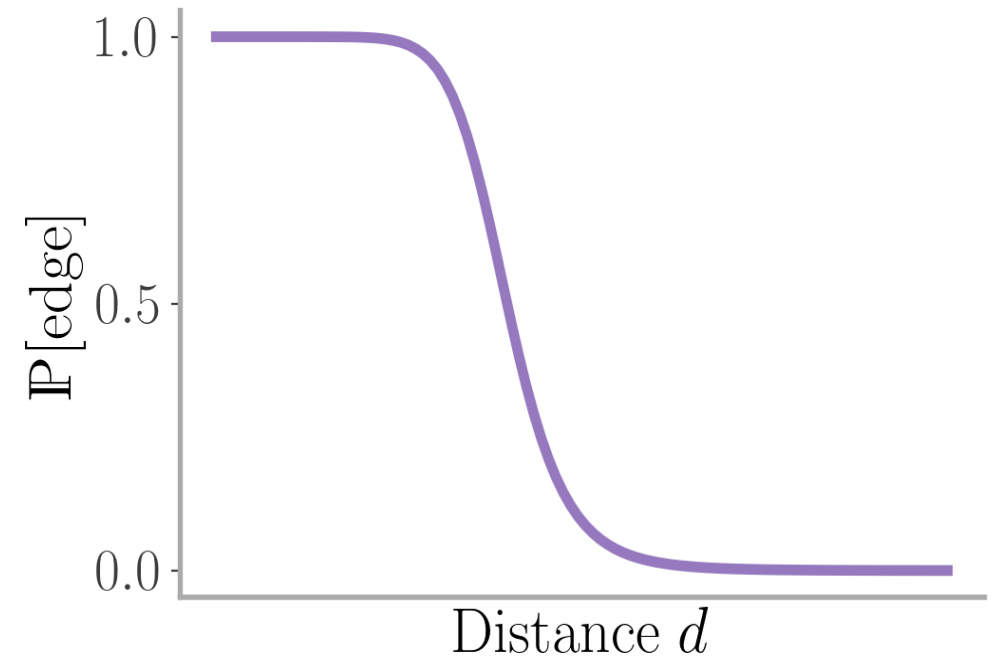
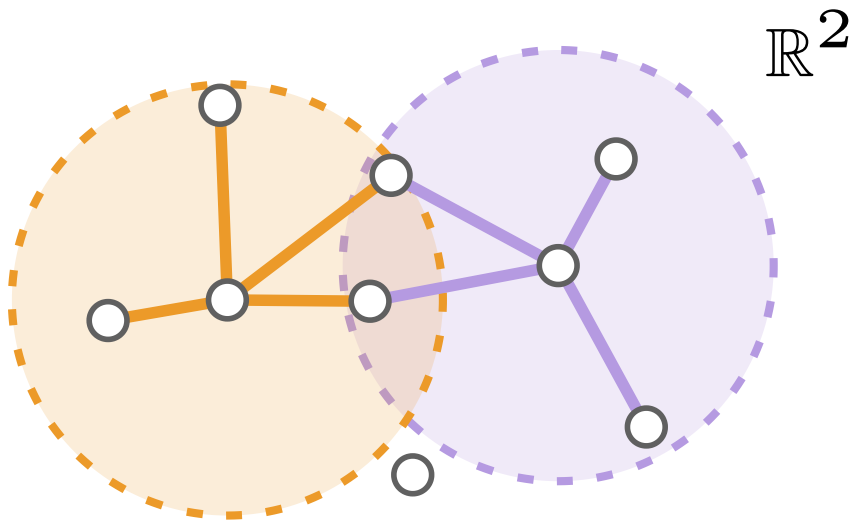
2025-05-06



Creating edges requires a cost

Creating edges requires a cost

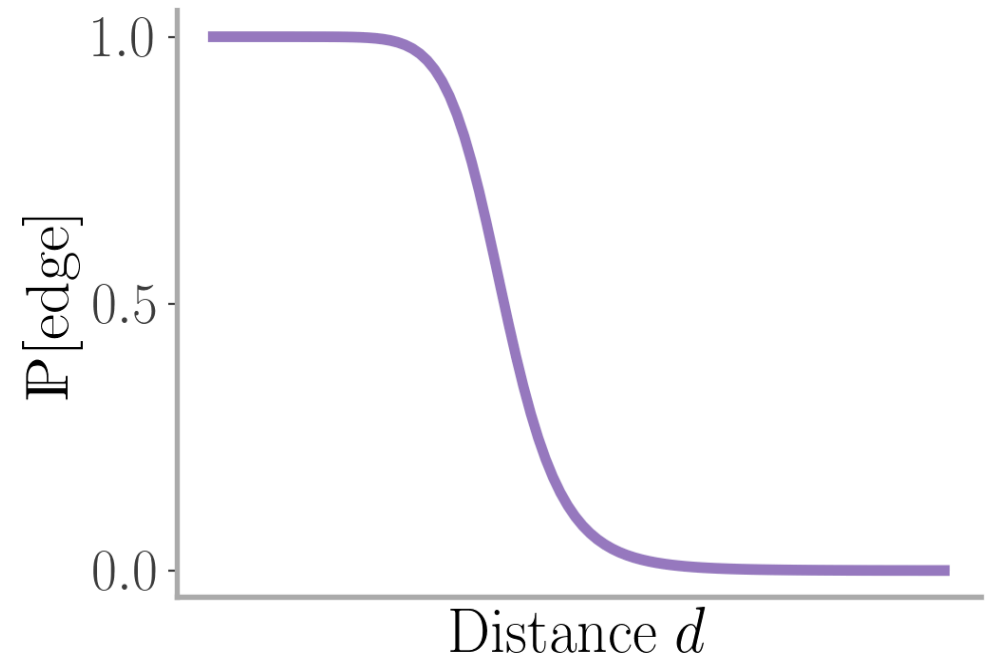
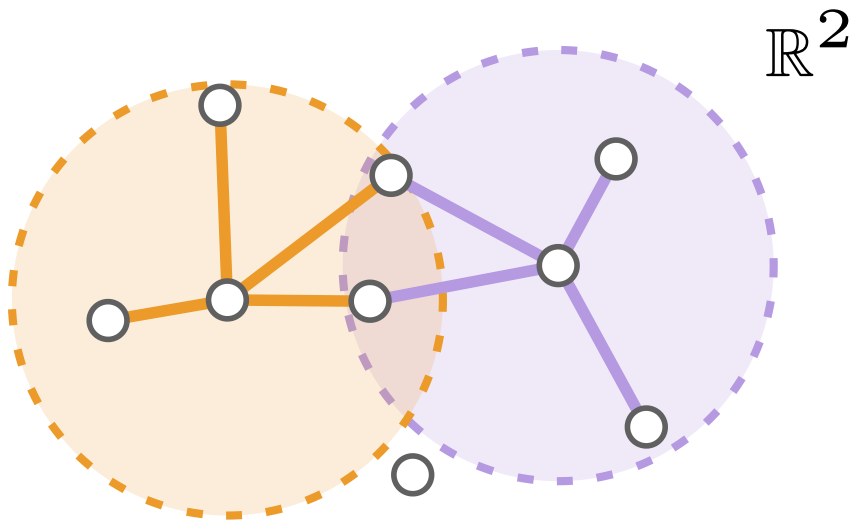
Vertices are placed in metric space. The edge cost **increases with its length**.



Creating edges requires a cost

Vertices are placed in metric space. The edge cost **increases with its length**.

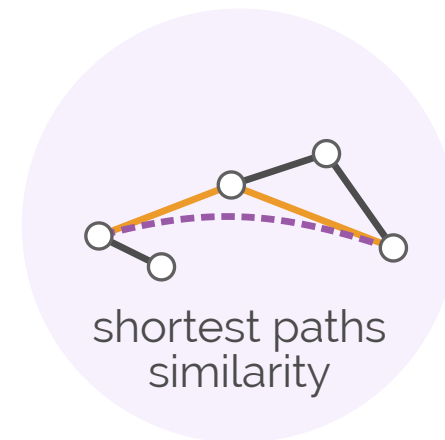
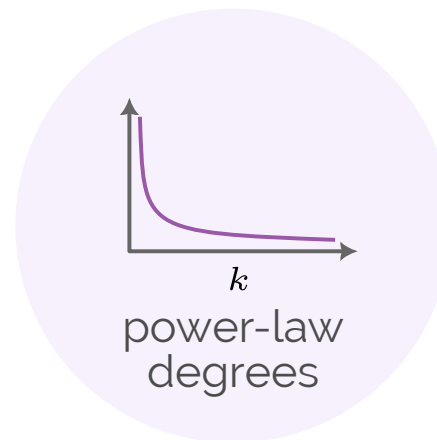
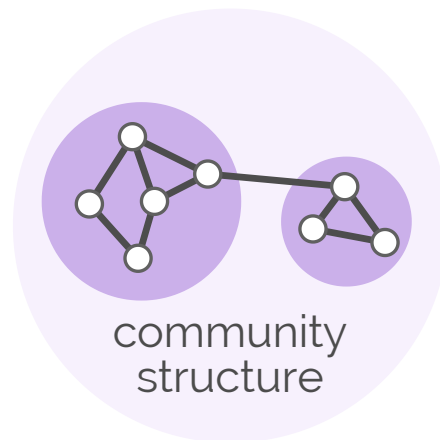
The metric space can be physical (e.g. transportation network, brain network) or not.



Latent hyperbolic space

Graphs obtained from hyperbolic space reproduce many empirically observed properties.

- Krioukov, D., et al. *Hyperbolic geometry of complex networks*. Phys. Rev. E **82**, 036106 (2010).
- Zuev, K., et al. *Emergence of Soft Communities from Geometric Preferential Attachment*. Sci. Rep. **5**, 9421 (2015).
- Krioukov, D., et al. *Clustering Implies Geometry in Networks*. Phys. Rev. Lett. **116**, 208302 (2016).
- Faqeeh, A., et al. *Characterizing the Analogy Between Hyperbolic Embedding and Community Structure of Complex Networks*. Phys. Rev. Lett. **121**, 098301 (2018).



... and others!

$\mathbb{H}^2$ and $\mathbb{S}^1$ random graph models

Each edge (u, v) exists with probability

$\mathbb{H}^2$ model

$$\mathbb{P}[(u, v) | x_u, x_v, \beta] = \frac{1}{1 + \exp\{\beta(d_{\mathbb{H}}(x_u, x_v) - R)\}},$$

where $d_{\mathbb{H}}$ is the hyperbolic distance, β is the sigmoid sharpness, R is the maximal radial coordinate, $x_u, x_v \in \mathbb{H}^2$ are the positions of vertices u and v respectively.

H^2 and S^1 random graph models

Each edge (u, v) exists with probability

H^2 model

$$P[(u, v) \mid x_u, x_v, \beta] = \frac{1}{1 + \exp \left\{ \beta(d_H(x_u, x_v) - R) \right\}},$$

where d_H is the hyperbolic distance, β is the sigmoid sharpness, R is the maximal radial coordinate, $x_u, x_v \in H^2$ are the positions of vertices u and v respectively.

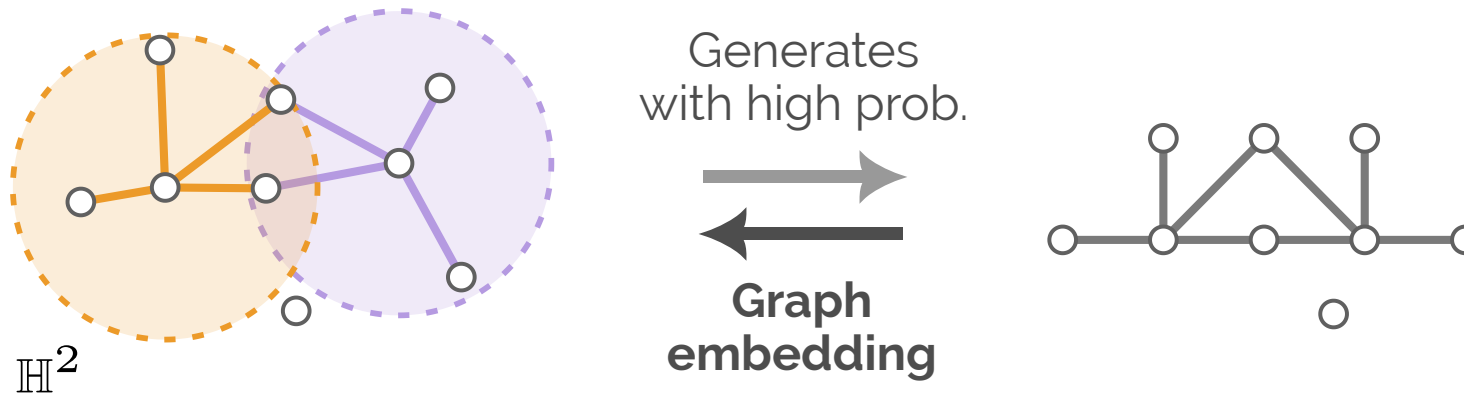
Using an approximation for d_H , this is equivalent to

S^1 model

$$p_{uv} = \frac{1}{1 + \left(\frac{d_S(\theta_u, \theta_v)}{\mu\kappa_u, \kappa_v} \right)^\beta} \approx P[(u, v) \mid x_u, x_v, \beta],$$

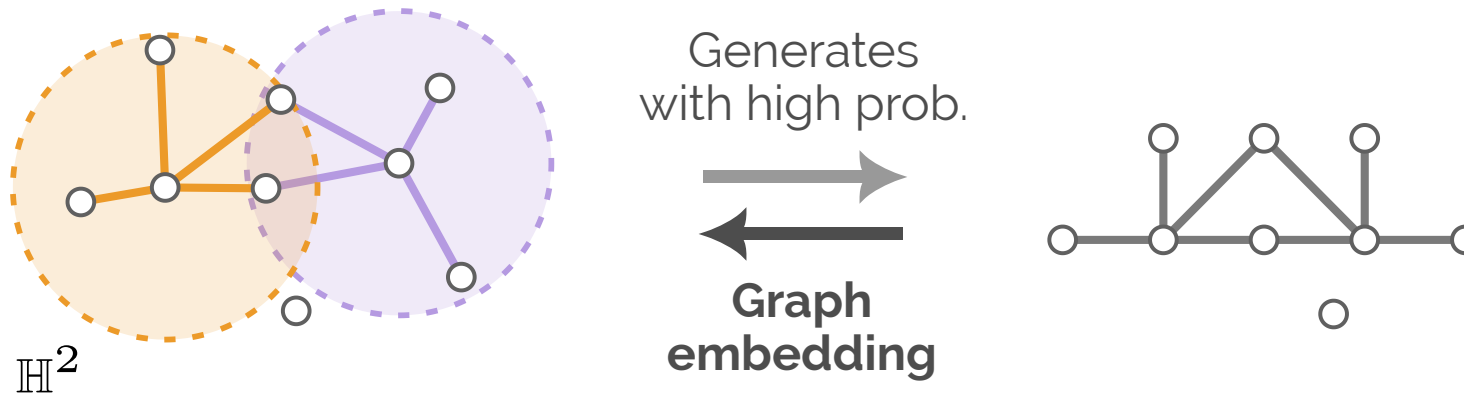
Graph vertex embedding in a nutshell

We want to **represent a given graph** using a hyperbolic embedding.



Graph vertex embedding in a nutshell

We want to **represent a given graph** using a hyperbolic embedding.



This amounts to:

Pairs of vertices are $\begin{cases} \text{close} & \text{if connected in graph,} \\ \text{far} & \text{if not connected in graph.} \end{cases}$

Embedding with maximum likelihood

The likelihood of obtaining a graph $G = (V, E)$ is simply

$$\mathbb{P}[G|\theta, \kappa, \beta] = \prod_{(u,v) \in V^2} p_{uv}^{a_{uv}} (1 - p_{uv})^{1-a_{uv}},$$

where $a_{uv} = 1$ if u and v are connected and is 0 otherwise.

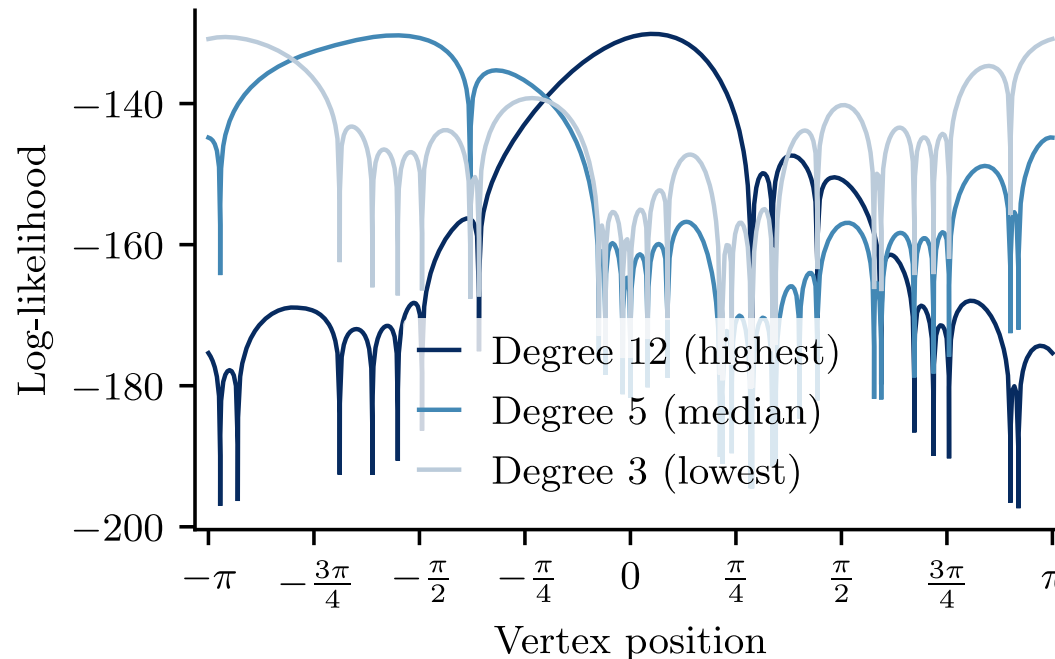
Embedding with maximum likelihood

The likelihood of obtaining a graph $G = (V, E)$ is simply

$$P[G \mid \theta, \kappa, \beta] = \prod_{(u,v) \in V^2} p_{uv}^{a_{uv}} (1 - p_{uv})^{1 - a_{uv}},$$

where $a_{uv} = 1$ if u and v are connected and is 0 otherwise.

Many algorithms give a maximum likelihood estimator (MLE). This is challenging because of the **abundance of local maxima**.



Current algorithms don't give the entire picture

Every algorithm

- Papadopoulos, F. et al. Phys. Rev. E **92**, 022807 (2015).
- Alanis-Lobato, G. et al. Appl. Netw. Sci. **1**, 1–14 (2016).
- Muscoloni, A. et al. Nat. Commun. **8**, 1615 (2017).
- García-Pérez, G. et al. New J. Phys. **21**, 123033 (2019).
- Wang, Z. et al. J. Stat. Mech.: Theory Exp. 123401 (2019).
- ...

yields a **single embedding**.

Current algorithms don't give the entire picture

Every algorithm

- Papadopoulos, F. et al. Phys. Rev. E **92**, 022807 (2015).
- Alanis-Lobato, G. et al. Appl. Netw. Sci. **1**, 1–14 (2016).
- Muscoloni, A. et al. Nat. Commun. **8**, 1615 (2017).
- García-Pérez, G. et al. New J. Phys. **21**, 123033 (2019).
- Wang, Z. et al. J. Stat. Mech.: Theory Exp. 123401 (2019).
- ...

yields a **single embedding**.

We currently ignore

- if there exists many plausible embeddings;
- how precise the vertex coordinates are.

We address both issues using a Bayesian approach.

$\mathbb{S}^1$ Bayesian model

The posterior of the Bayesian $\mathbb{S}^1$ model is

$$p(\theta, \kappa, \beta | G) \propto \mathbb{P}[G | \theta, \kappa, \beta] p(\beta) \prod_{v \in V} p(\theta_v) p(\kappa_v),$$

where the priors are

$$\begin{aligned} \theta_u &\sim \text{Uniform}[-\pi, \pi), \\ \kappa_u &\sim \text{Cauchy}, \quad \kappa_u > \epsilon, \\ \beta &\sim \text{Normal}, \quad \beta > 1. \end{aligned}$$

S¹ Bayesian model

The posterior of the Bayesian S¹ model is

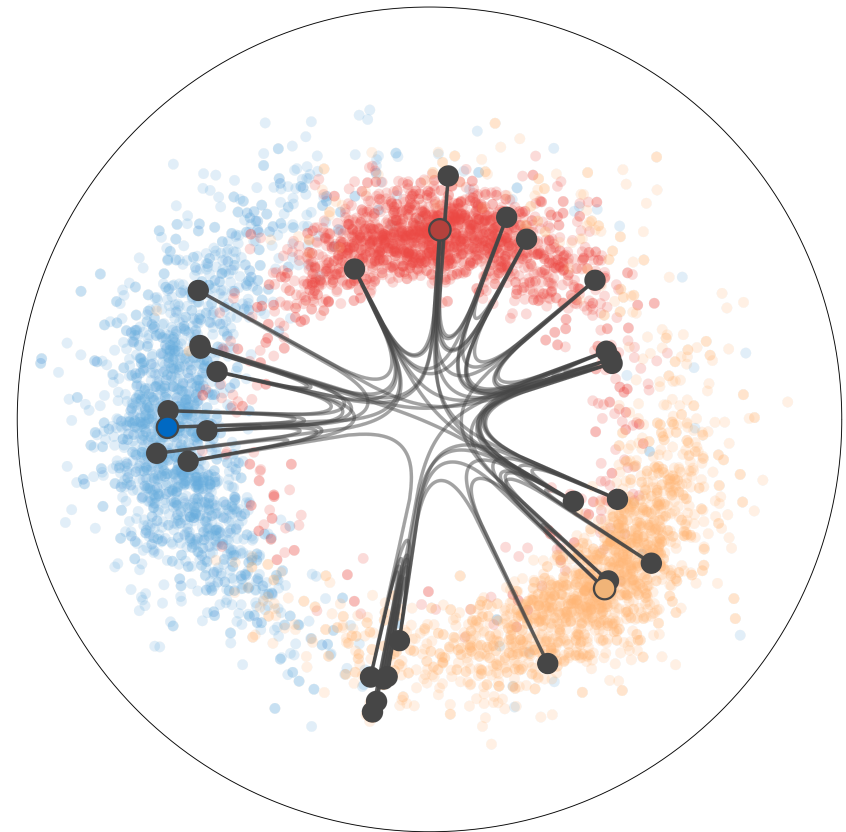
$$p(\theta, \kappa, \beta | G) \propto P[G | \theta, \kappa, \beta] p(\beta) \prod_{v \in V} p(\theta_v) p(\kappa_v),$$

where the priors are

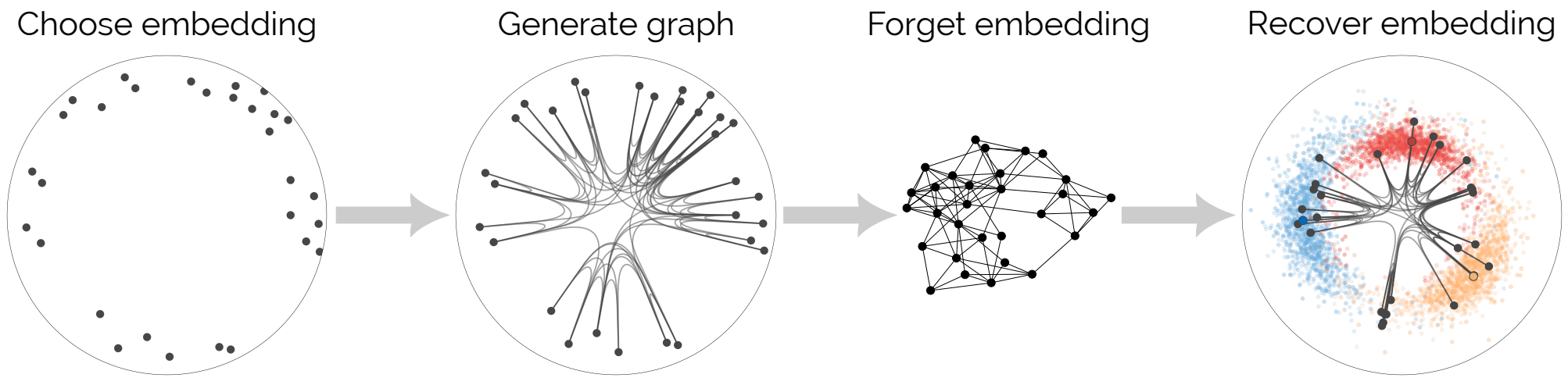
$$\theta_u \sim \text{Uniform}[-\pi, \pi),$$

$$\kappa_u \sim \text{Cauchy}, \quad \kappa_u > \epsilon,$$

$$\beta \sim \text{Normal}, \quad \beta > 1.$$

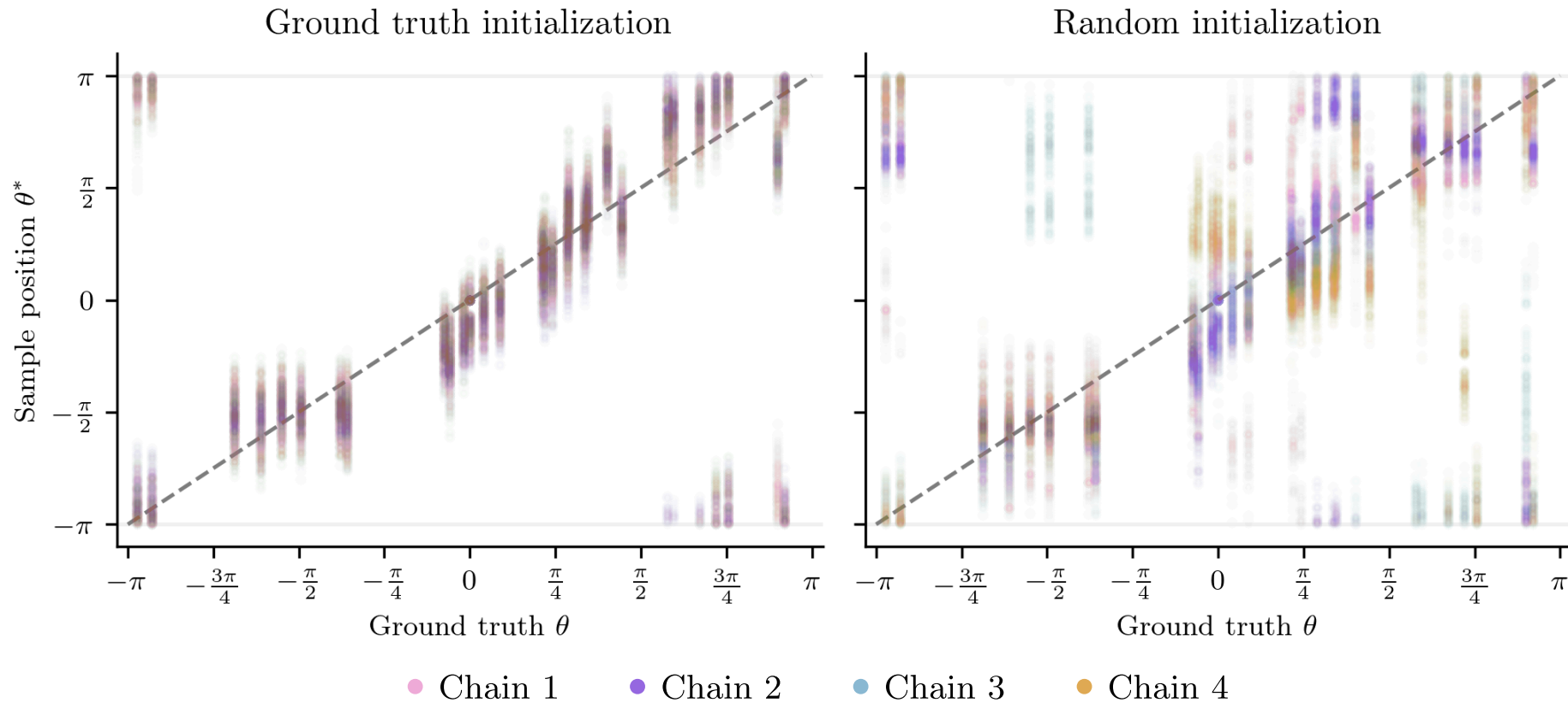


Sanity check with synthetic data



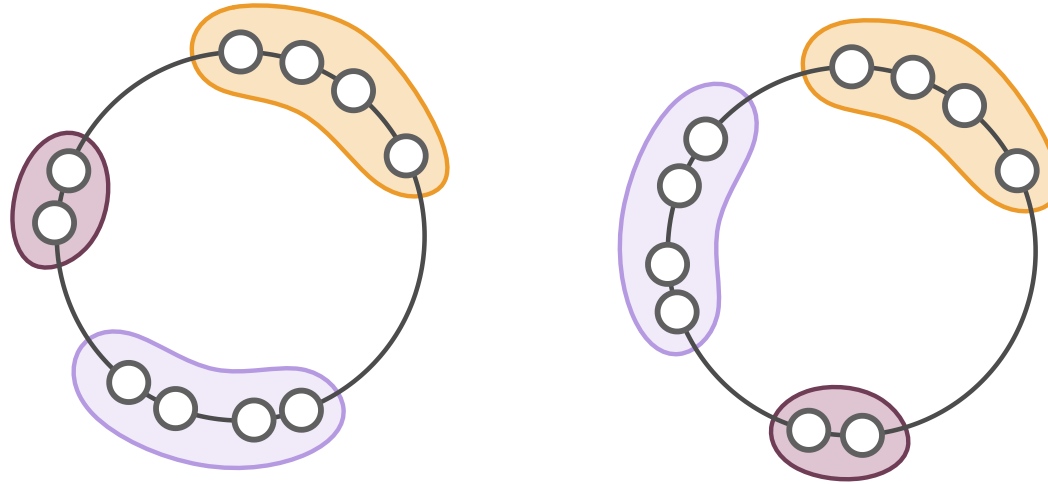
Usual sampling methods don't work

Hamiltonian Monte Carlo¹ (HMC) and random walk don't sample properly.



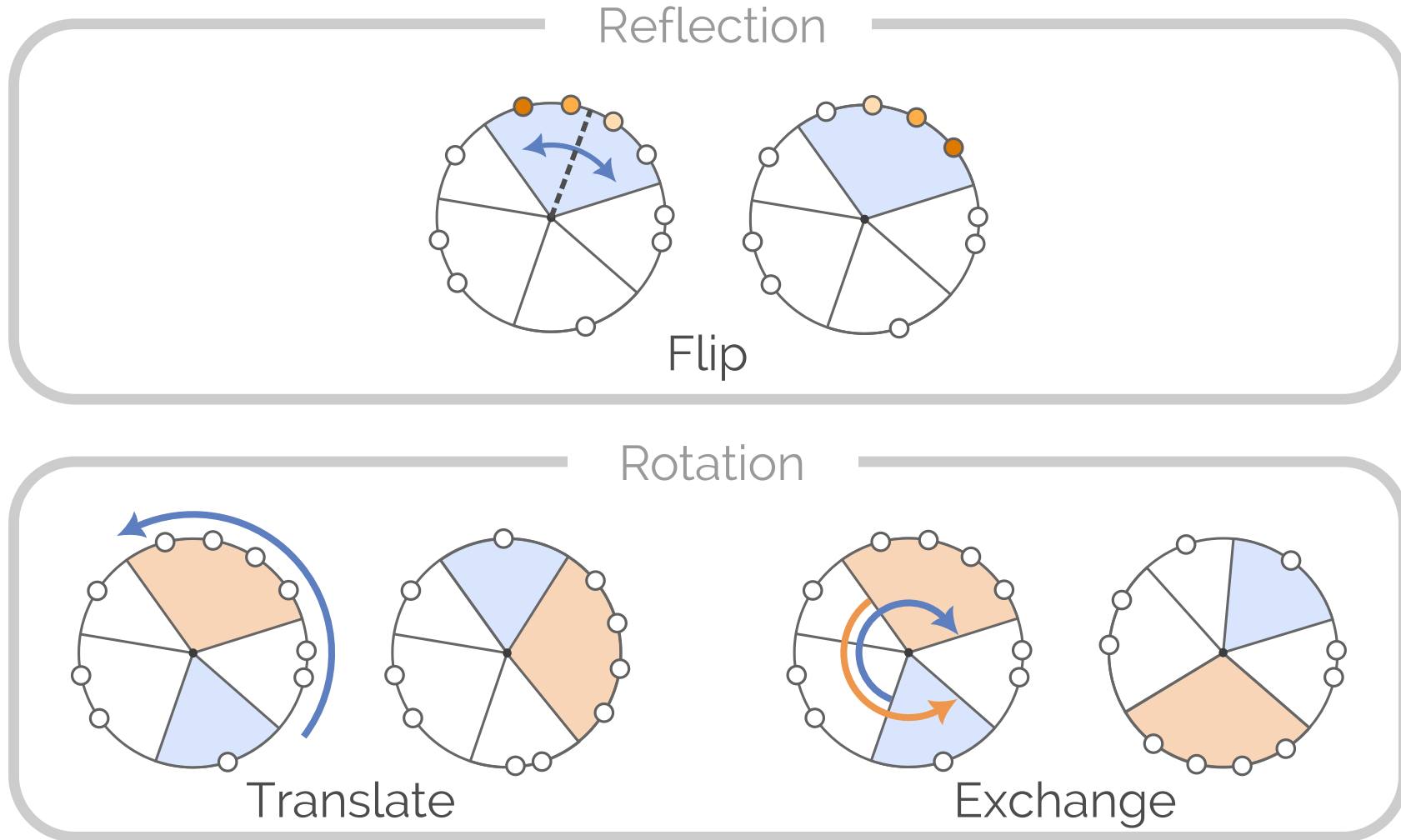
Locality implies clusters

Since edges are local, groups of nearby vertices should be moved together.



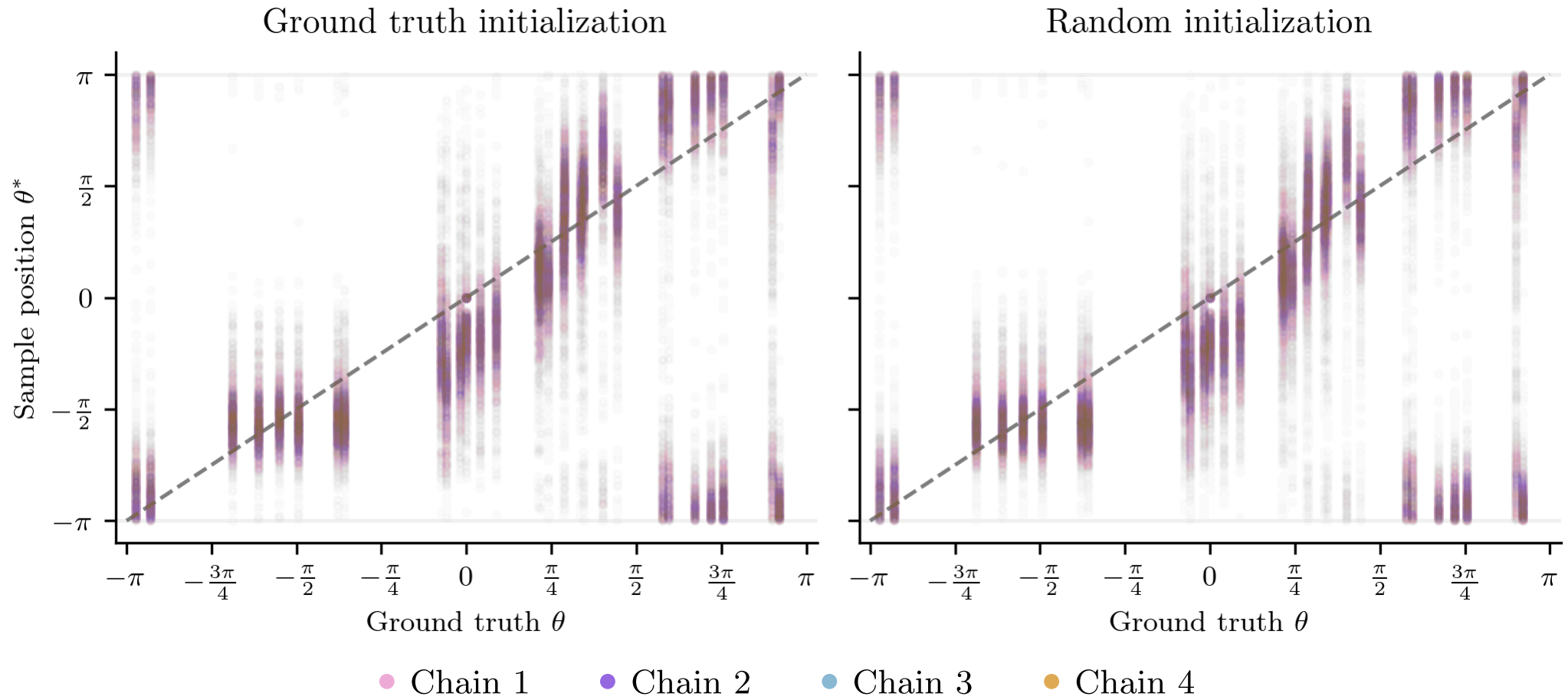
Locality implies clusters

Since edges are local, groups of nearby vertices should be moved together.

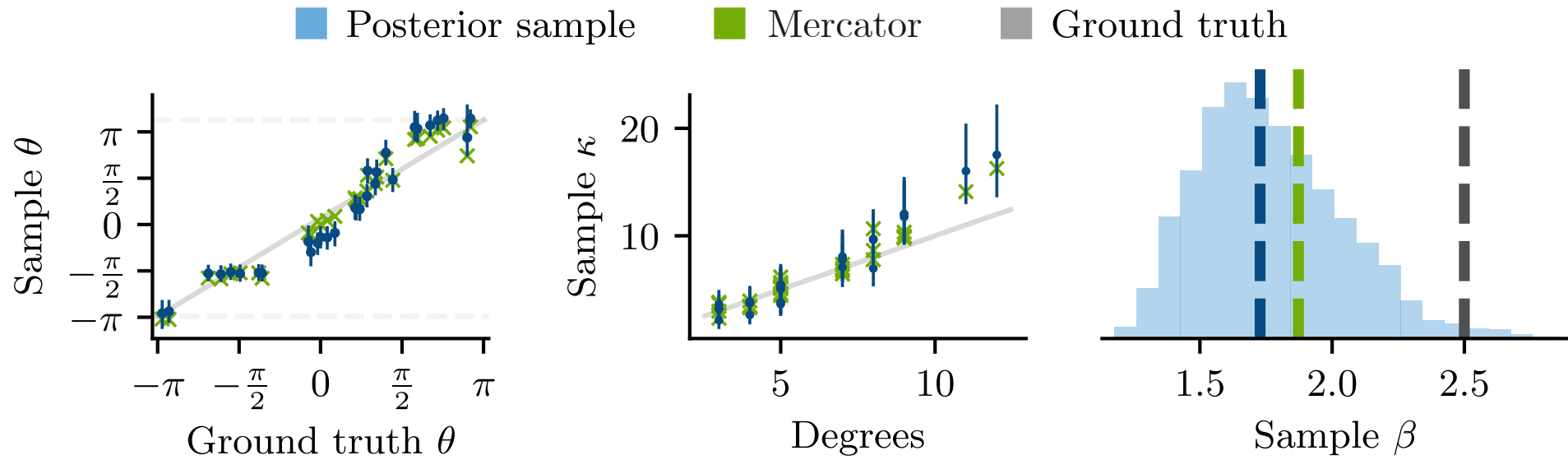


Cluster transformations fix the sampling issue

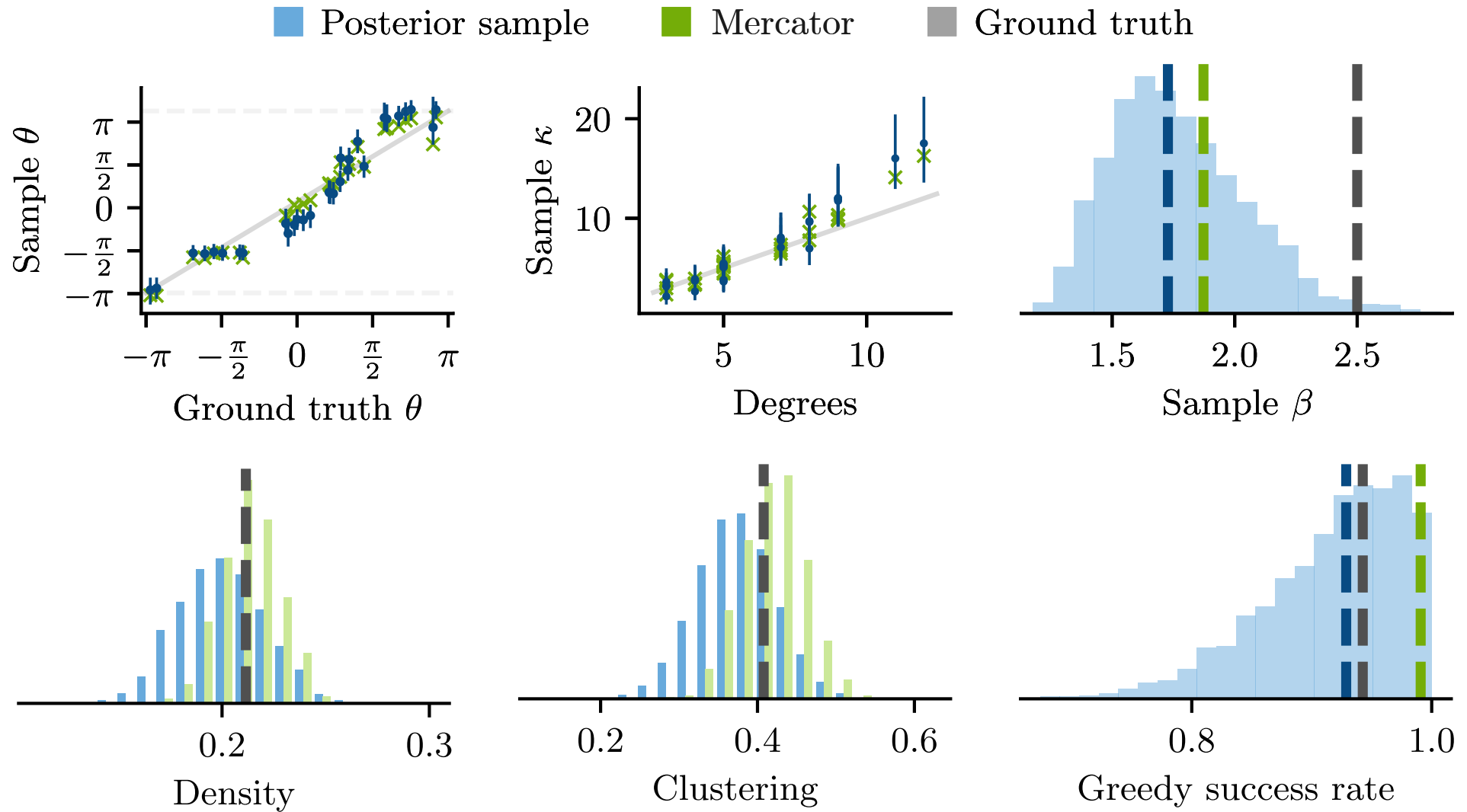
Cluster transformations + random walk yield good samples of the posterior.



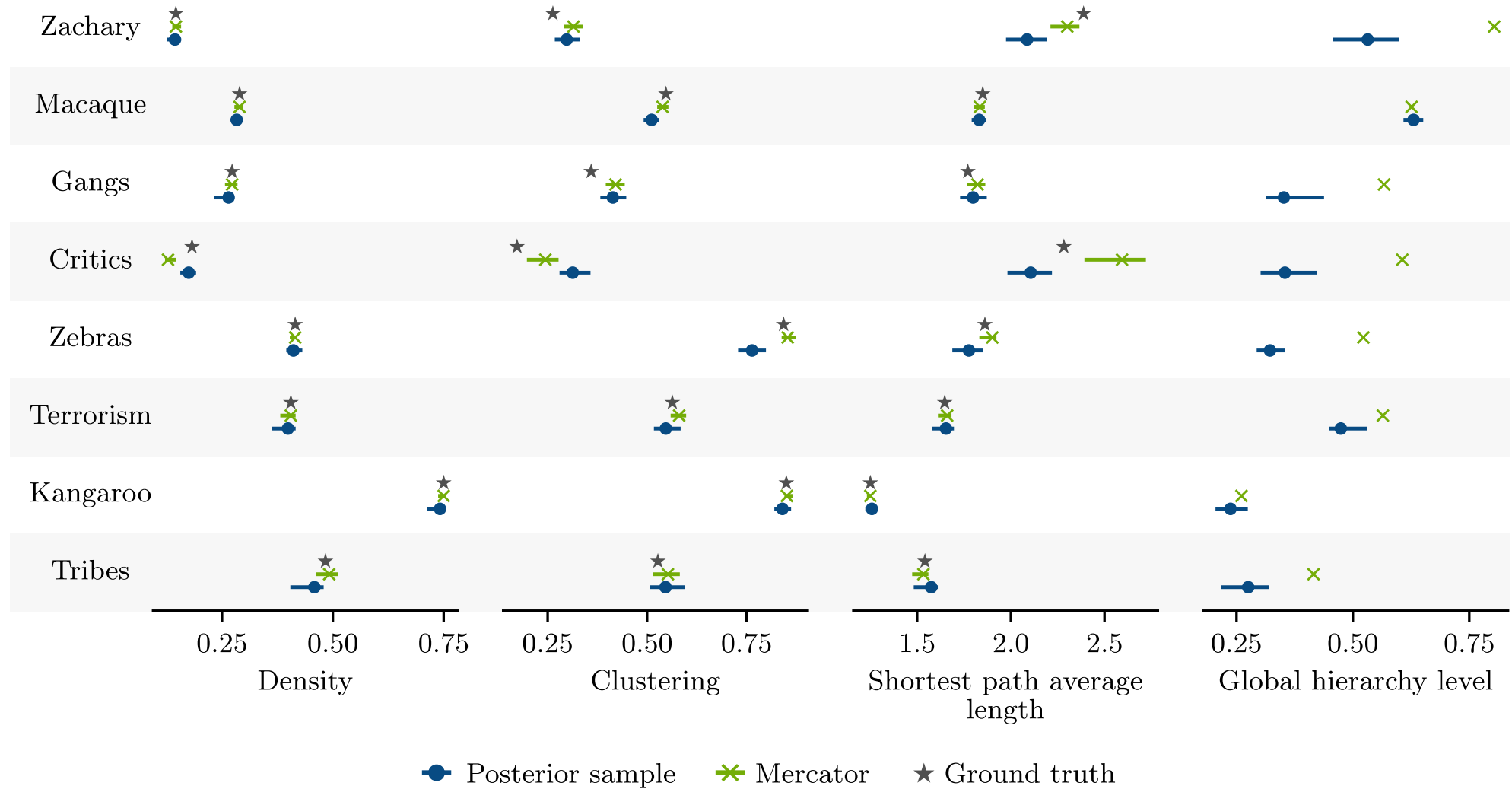
Embedding error bars



Embedding error bars



Empirical networks properties



Induced multimodal distribution

Conflicting ground truth model:

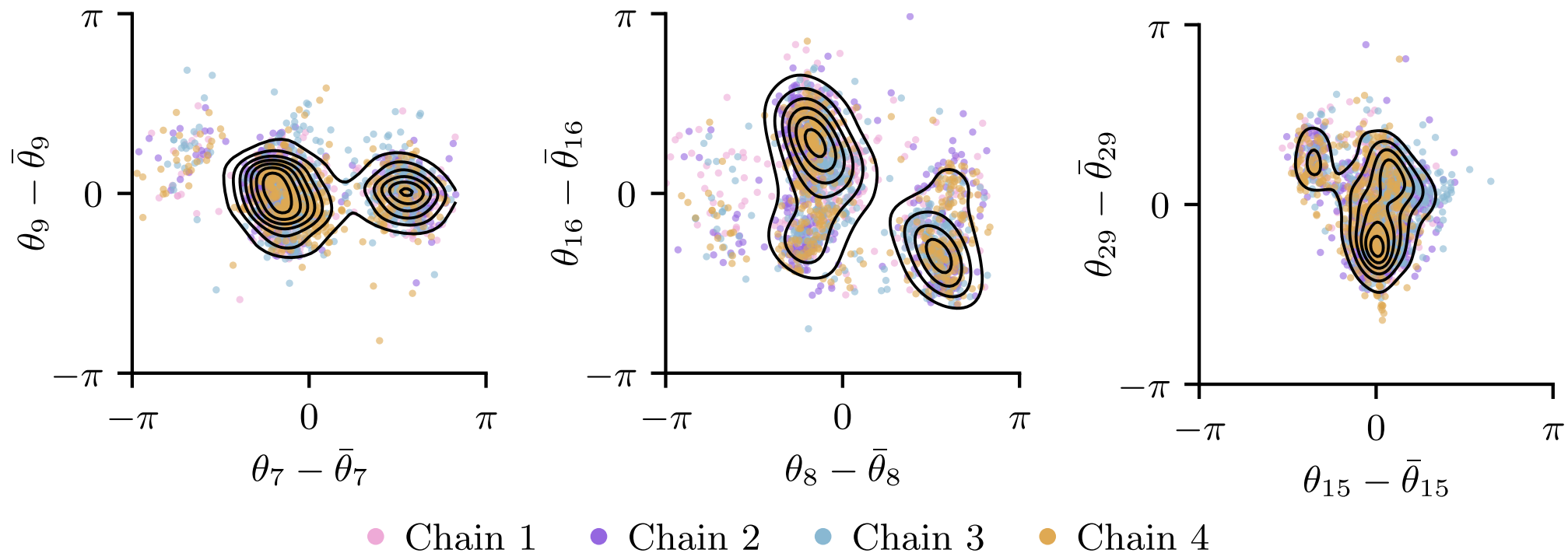
- A vertex v is given two positions $\theta_v^{(1)}$ and $\theta_v^{(2)}$.
- When generating G with the $\mathbb{S}^1$ model, each edge probability including v uses randomly $\theta_v^{(1)}$ or $\theta_v^{(2)}$.

Induced multimodal distribution

Conflicting ground truth model:

- A vertex v is given two positions $\theta_v^{(1)}$ and $\theta_v^{(2)}$.
- When generating G with the S^1 model, each edge probability including v uses randomly $\theta_v^{(1)}$ or $\theta_v^{(2)}$.

Marginal posterior distributions



Takeaways

- Hyperbolic random geometric graphs reproduce many empirically observed properties.
- Locality $\implies$ clusters as coarse-graining;
- Bayesian approach finds error bars and can identify multiple good embeddings.

Paper:

Lizotte, A., Young, J.-G. and Allard A. *Symmetry-driven embedding of networks in hyperbolic space*. arXiv:2406.10711 (2024).

[Accepted at Communication Physics]



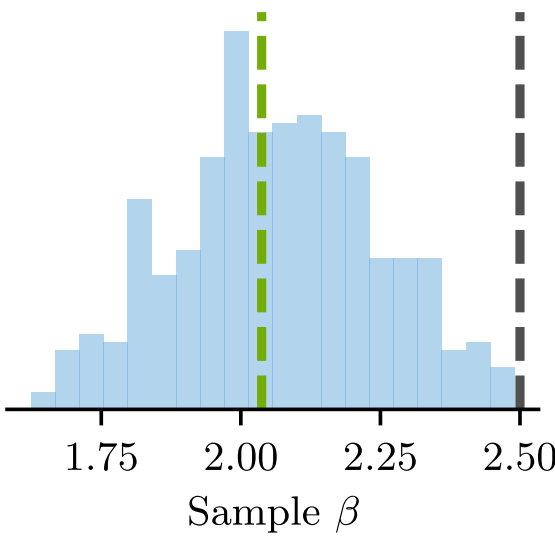
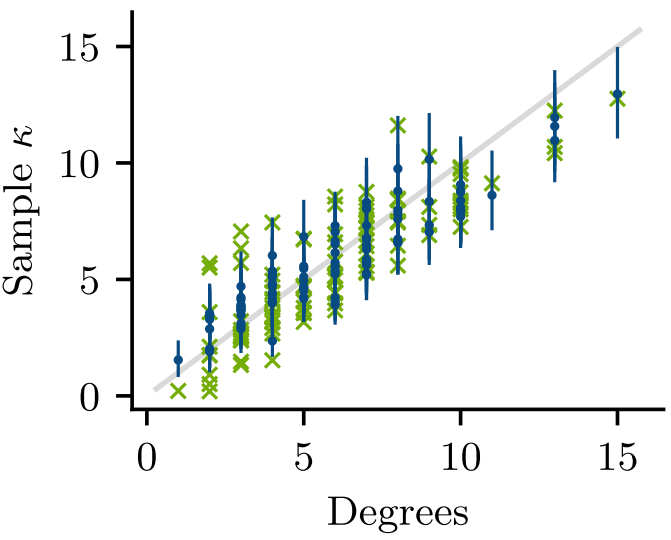
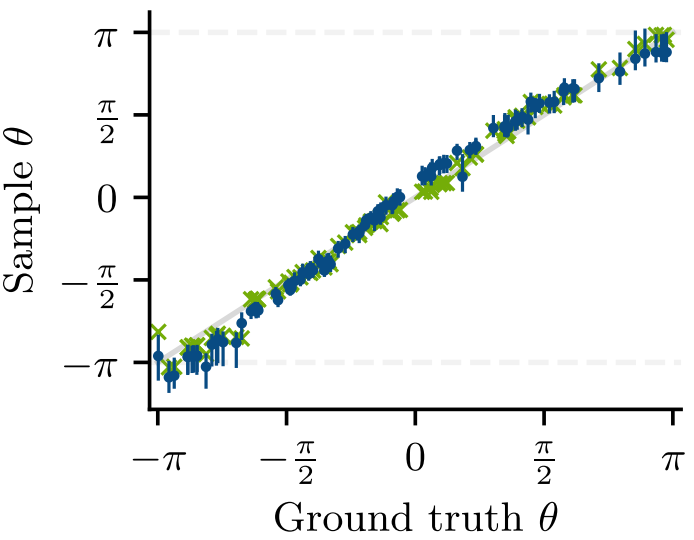
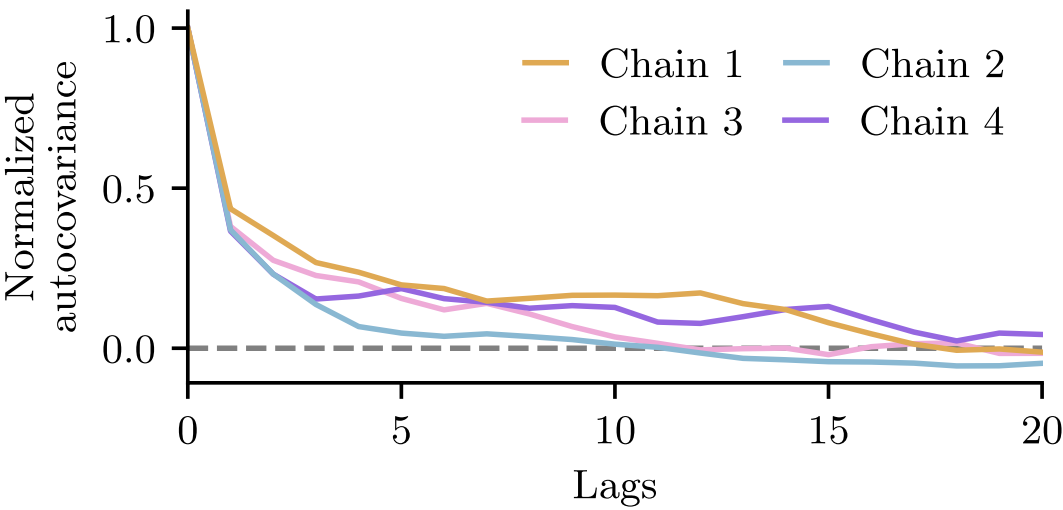
Antoine Allard



Jean-Gabriel Young

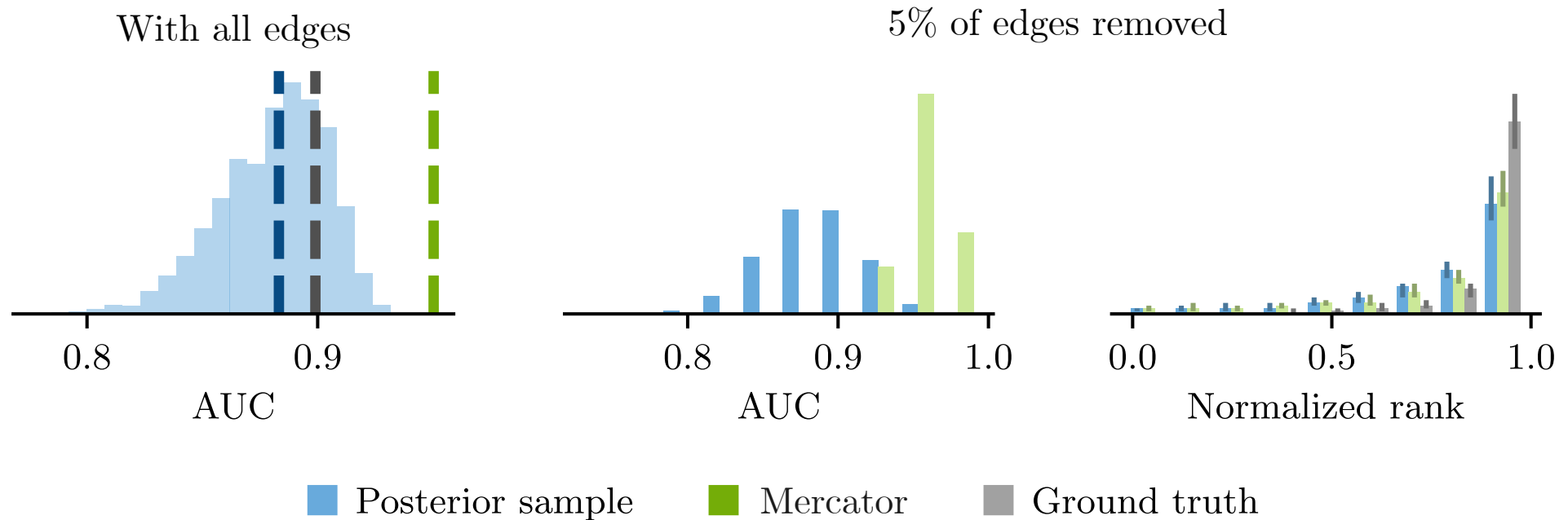


Synthetic graph of 100 vertices

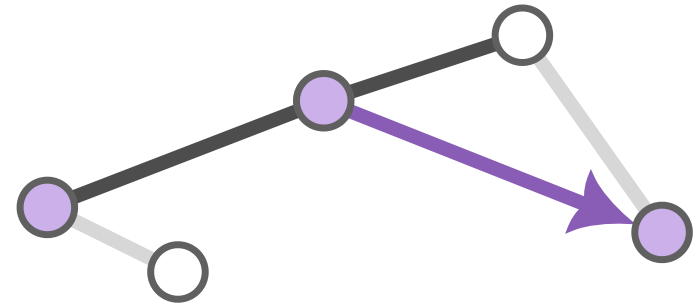
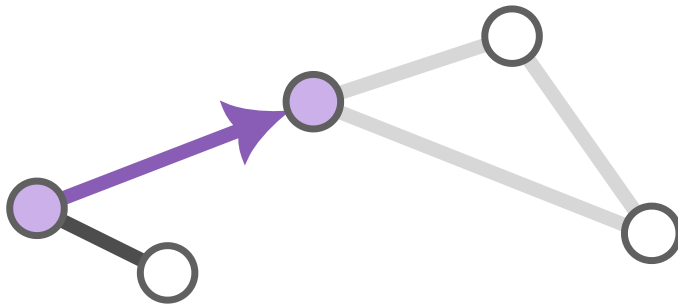
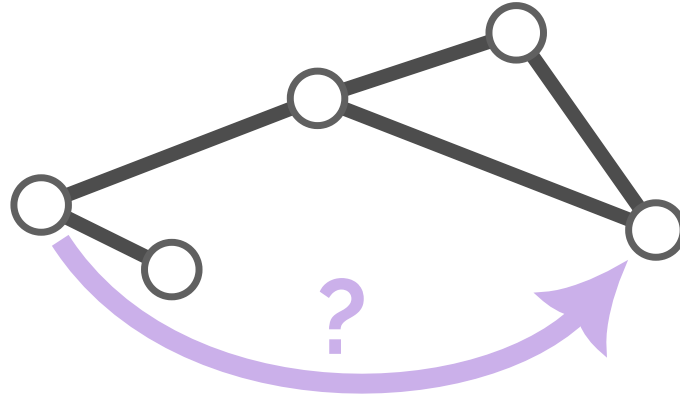


■ Posterior sample ■ Mercator ■ Ground truth

Link prediction is equivalent when sampling



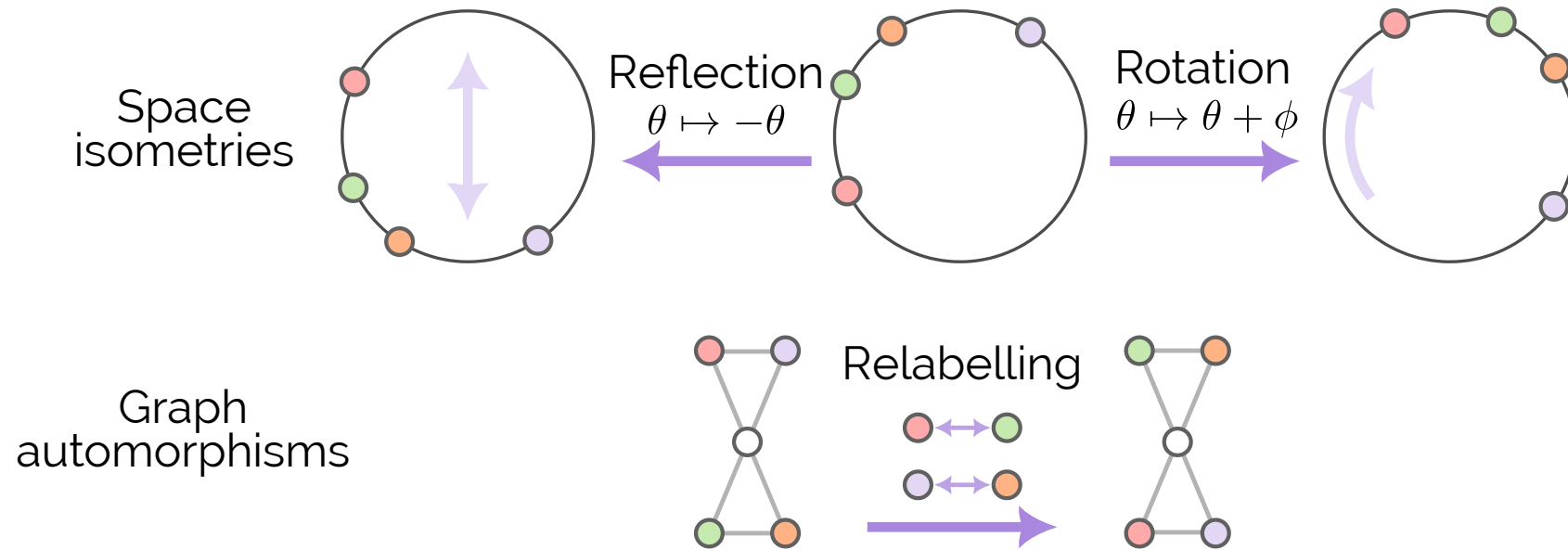
Greedy routing



Go to the neighbour closest to the destination.

Model symmetries

The $\mathbb{S}^1$ model is not identifiable because of graph and space symmetries.



Comparing embeddings requires alignment.